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Code JupyterLab Python [conda env:bas

Introduction to Classification (Machine Learning)

What is Classification?

Classification is a supervised learning technique in Machine learning whare the goal is to predict to a category or label for a given input data.

Key Idea

- Instead of predicting a continuous value(like salary or house price in regression),classification predicts discreate output(classes)

Examples of Classification Problems

- Yes/ No
- Spam / Not spam
- Pass / Fail
- Disease / No disease
- Cat/ Dog classification
- Approved/ Rejected loan

Logistics Regression

logistic Regression is a supervised learning algorithm used for classification problems.

It is mainly used when the output variable is binary (two classes),such as:

- Yes/ No
- Pass / Fail

- Pass / Fail
- 0/1
- Spam/Not spam
- In Logistic regression does not predict continuous values.
- Instead,it predicts the probability of a class.

Example

If we want to predict whether a student will pass:

- Input: study hours,attendance,marks
- Output: probability of passing(0-1)

Then we convert probability to class:

- if probability > 0.5 - pass
- if probability < 0.5 - fail

Hypothesis Function(Linear Part)

First, Logistic Regression computes a linear combination of inputs: z = wX + b where

- w=weight
- X=input feature
- b=bias
- z=linear output

Sigmoid Function

To convert the linear output into a probability ,Logistic Regression uses the Sigmoid Function.

properties

- Output is always between 0 and 1
- Converts any real number into probability
- Produces an S-shaped (Sigmoid) curve

Probability Interpretation

After applying the sigmoid function ,we get probability.

Interpretation

```
* if probability >= 0.5 -> class = 1
* if probability < 0.5 -> class=0
```

Decision Boundary ¶

The decision bounday is the threshold that separates the two classes in Logistics Regression.

Condition

```
The decision boundary occurs when :
                wX + b =0
```

Meaning

Meaning

- One side of the boundary -> class 0
- Other side of the boundary -> class 1

[4]:
```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression

# Load dataset
df = pd.read_csv("student_data.csv")

# Features (X) and Target (y)
X = df[['Hours_Studied']]
y = df['Pass']

# Train-test split

X_train, X_test, y_train, y_test = train_test_split(

    X, y, test_size=0.2, random_state=42

)

# Create model
model = LogisticRegression()

# Train model
model.fit(X_train, y_train)

# Predictions
y_pred = model.predict(X_test)
```

```
# Create model
model = LogisticRegression()

# Train model
model.fit(X_train, y_train)

# Predictions
y_pred = model.predict(X_test)

# Predict new student
new_student = pd.DataFrame({

    'Hours_Studied': [2]

})
prediction = model.predict(new_student)
print(y_pred)
print("Prediction (0=Fail, 1=Pass):", prediction[0])
```

```
[1 0]
Prediction (0=Fail, 1=Pass): 0
```

K-Nearest neighbours(KKN)

- K-Nearest neighbours(KKN) is a supervised machine learning Algorithm used for:
    - Classification
    - Regression
- KNN is most commonly used for Classification problem.
- The algorithm predicts the class the class of a new data type point by looking at the

Why it is called K-Nearest Neighbours?

K

- The number of nearest neighbours to consider Nearest
- The closest data points based on the distance Neighbours
- The existing data points smallest distance - more similar points largest mdistance - less similar points

Choosing the value k

the value of k plays a crucial role in the performance of the KNN algorithm

Distance Metric

- KNN identifies the nearest neighbors by calculationg the distance betweeen data points.
- the most commonly used distance metrics is euclidean distance.

where:

- d = distance between two points.
- (x1,y1) = First point
- (x2,y2) = Second point

Simple explanation

- Eucliden distance measures the

Interpretation

- The dstance between point A and Point B is 5 units

Interpretation

- The dstance between point A and Point B is 5 units
- In KNN:
  - smaller distance = more similar points
  - larger distance = less similar points.

Why distance is important in KNN:

- KNN predicts a class by finding the nearest neighbors.
- The distance metric helps determine :
  - Which point are closest.
  - Which neighbors should participate in voting.

Choosing the Value of K

- The value of K plays a crutial role in the performance of the KNN algorithm.
- Choosing a very small or very large value of K can negatively affect prediction accuracy.

Case 1: Small K

Example

```
* K = 1
```

- The model looks at the only one nearest neighbor .

Advantage

- Captures local patterns effectively.

Advantage

- Captures local patterns effectively.
- Can model complex decision boundaries

Disadvantages

- Very sensitive to noise and outliers.
- A single incorrectly labeled point can lead to a wrong prediction .

Example

- Suppose the nearest neighbor is mistakenly as pass instead of fail.
- Since K =1, the mode directly predicts Pass resulting

Case 2 : Large K

Example:

- k = 10
- The model consisders many neighors before making a prediction.

Advantages:

- More stable predictions.
- Less affected by noise data.

Disadvantages:

- Important local patterns may be ignored.
- Different classes may get mixed together.

Different classes may get mixed together.

Example:

- If most nearby students failed but many distant students passes , a large K may incorrectly predict Pass.

Visual Understanding

Small K	Large K
Sensitive to noise	Stable predictions
Complex decision boundary	Smooth decision boundary
Overfitting risk	Underfitting risk

Case 3 : Best Choice of K

A common rule of thump is

where:

K = Number of neighbors N = Number of training samples.

Example:

Given:

```
* N=100
```

Curse of Dimensionality in KNN

What is the curse of dimensionality ?

- The curse of dimensionality refers to the problems that occue when the number of features (dimensions) in a dataset becomes very large.
- KNN relief on distance calculaitons to find the nearest neighbors . As the number of dimensions increases , distance calculations become less effective , making it difficult for KNN to identify truly similar data type.

What is dimension ?

Number of dimensions = 2

Example 2

Sample Dataset

Hours	Attendance	Age	Marks	Income	Salary
5	80	20	75	20000	30000

Number of dimensions = 6

What happens as dimensions increase ?

As the number of features grows:

- data points become sparse(spread far apart).

Distance calculations become less meaningful.

- Nearest neighbors become harder to identify.

- Nearest neighbors become harder to identify.
- Model accuracy may decrease.

Why is it a Problem?

KNN works by finding the nearest neighbors using distance .

Low dimensions(2 or 3 features) Data points are relatively

nearest neighbor are easily identify

Effects on KNN

1. Reduce Accuracy

- The identified nearest neighbors may not actually are similar

2. Increased Computation

- More features mean more distance calculations

3. Higher Memory Usage

- KNN stores the entire trainng dataset.

Solutions

Definiton

Feature selection

Dimensionality Reduction

Common tech

```
* Principal Component Analysis(PCA)
* Principal Component Analysis(PCA)
* Linear Discriminant Analysis(LDA)
```

These techniques compress data into fewer dimensions.

What is decision Tree?

- A Decision Tree is a supervised machine learning algorithm used for :
  - Classification
  - Regression
- It works like a flowchart , where each decision leads to another question until a final prediction is reached.

Does a decision tree work ?

- A decision tree learns by asking a series of questions about the data.

At each step :

1. Select the best feature for spelling the data.
2. Divide the data into smaller groups .
3. Repeat the process on each group.
4. Stop when a prediction can be made.

- This process is called recursive spliting.

Components of Decision Tree

Components of Decision Tree

1. Root Node:

- The top most node of the tree.

Example:

- Study hours > 5?
- This is the first question asked.

2. Decision Node

- A node where a decision or condition is checked.

Example:

```
* Attendance > 75?
```

3. Leaf Node

- The final output or prediction.

Example:

Pass or fail

Gini Impurity

- Gini impurity is a measure used in decision trees to determine how mixed the classes are within a node.

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Code

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Gini Impurity

- Gini impurity is a measure used in decision trees to determine how mixed the classes are within a node.
- It tells us the probability of incorrectly classifying a randomly chosen sample if we assign a class label accordingly to the class distribution in that node.
- Think of a ox contating students labeled as Pass or Fail.

Case 1 : Mixed Classes

Result Summary

Result	Count
Pass	5
Fail	5

- Total students = 10

- If randomly pick a student , it is difficult to predict whether the student elongs to the pass or fail class because both classes appear equally .

Case 2 : Pure classes

Result Summary ¶

Result	Count
Pass	10
Fail	0

[ ]: